

Research Progress Report

XY and Heisenberg Spin Models with Magnetic Anisotropy

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1. Research Direction Following the Ising Model

The previous report used a 2D Ising model to study hysteresis, coercivity, and the effect of temperature. One of the main limitations of that model is that each spin is restricted to only two possible states, +1 or -1. Real magnetic moments can rotate continuously, so the next step was to extend the simulation to continuous-spin models that can include a controllable magnetic anisotropy.

Following this direction, I implemented an XY model and began extending the same framework to a Heisenberg model. The main question is how the hysteresis curve changes as the anisotropy strength D is increased, and whether the continuous spin models approach the Ising model behavior as D approaches infinity.

2. XY Model with Anisotropy

In the XY model, each spin can point in any direction within the XY plane rather than being restricted to two states. Each spin is represented by a single angle θ_i :

$$\mathbf{S}_i = (\cos \theta_i, \sin \theta_i)$$

For a magnetic field applied along the x direction, the energy used in the simulation can be written as:

$$E = -J \sum \langle i, j \rangle \cos(\theta_i - \theta_j) + D \sum_i \sin^2 \theta_i - H \sum_i \cos \theta_i$$

The Metropolis procedure is similar to the Ising model, but instead of proposing a complete spin flip, the update proposes a new spin angle. The energy before and after the proposed rotation is compared, and the change is accepted using the same Metropolis criterion.

To test the simulation works, I ran 1000 sweeps with the initial conditions: $J=1$, $T=10$, $D=100$, $H=0$.

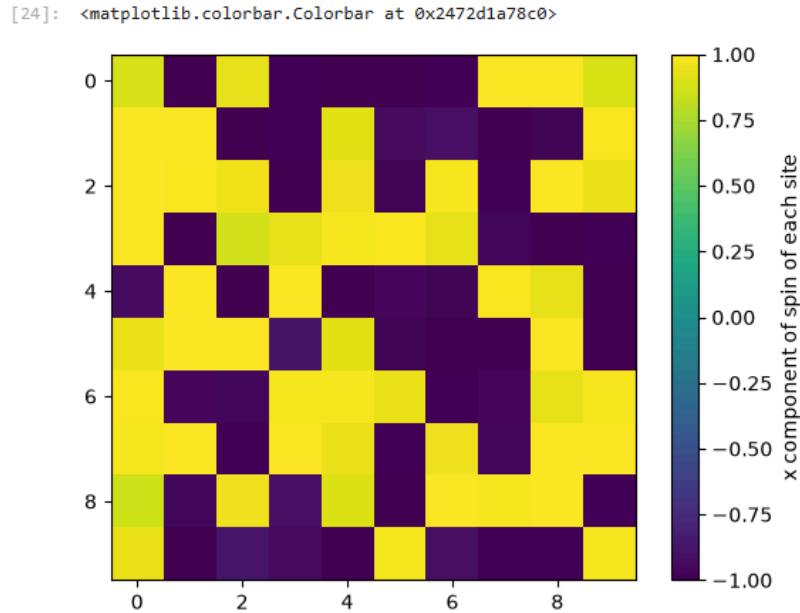


Figure1: the x-component of spins in the 10x10 lattice after 1000 Monte Carlo sweeps with initial conditions: $J=1$, $T=10$, $D=100$, $H=0$

The figure shows that even at the relatively high temperature of $T = 10$, most spins remain strongly aligned close to either the $+x$ or $-x$ direction when $D = 100$. The x-components are therefore concentrated near +1 and -1 rather than taking a wide range of intermediate values. This demonstrates the effect of strong anisotropy, which

energetically confines the spins toward the preferred x-axis despite thermal fluctuations. The presence of both positive and negative regions is expected because $H = 0$, so neither $+x$ nor $-x$ is favored by the external field. At sufficiently large D , this behavior begins to resemble the two-state nature of the Ising model.

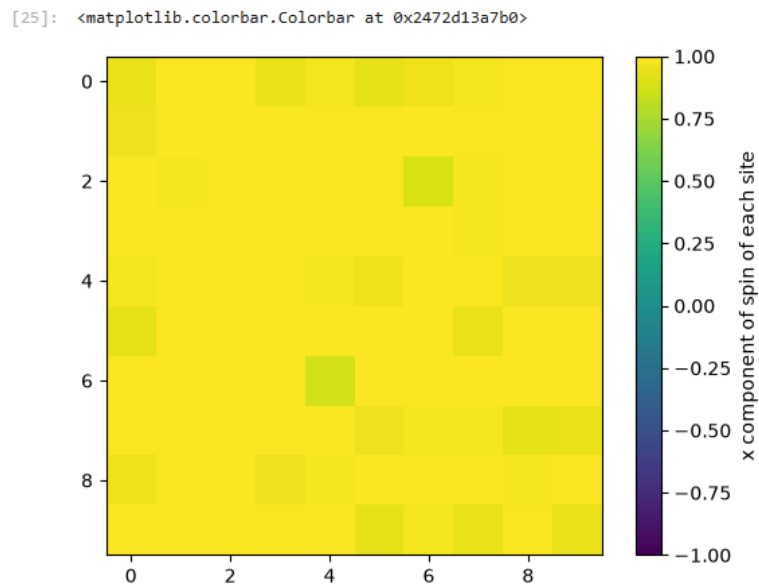
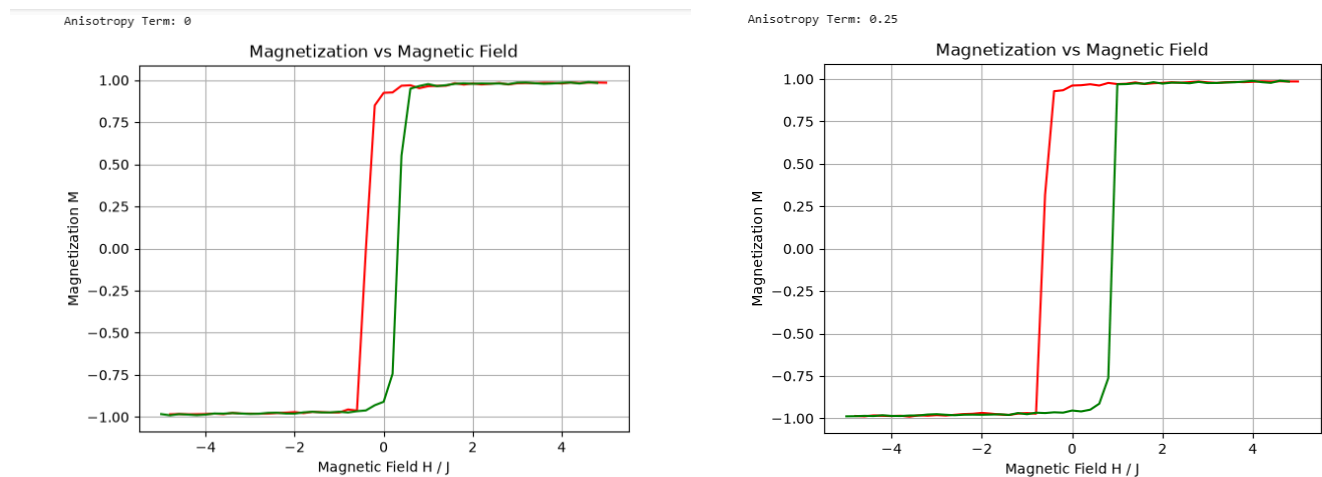


Figure2: the x-component of spins in the 10x10 lattice after 1000 Monte Carlo sweeps with initial conditions $J=1$, $T=10$, $D=100$, $H=100$.

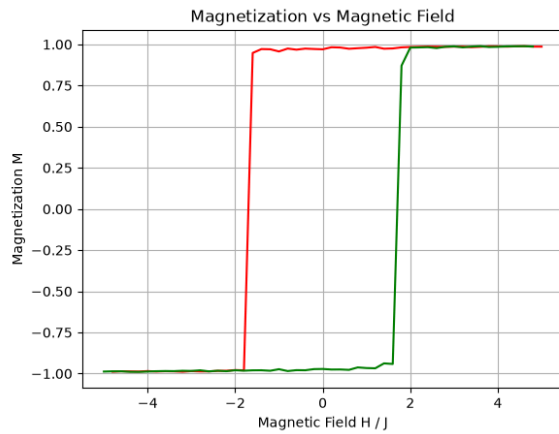
Now after setting $H = 100$, the figure shows that nearly all spins become aligned in the $+x$ direction, with x-components close to $+1$. The anisotropy term $D = 100$ already confines the spins close to the x-axis, but by itself it does not favor $+x$ over $-x$. The positive external field breaks this symmetry and lowers the energy of spins pointing along $+x$, causing the lattice to approach a fully magnetized state. The small deviations from $+1$ are due to thermal fluctuations at $T = 10$.

3. Effect of Anisotropy on XY Hysteresis

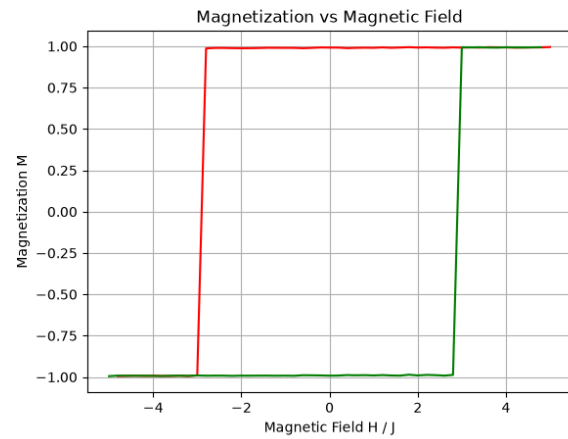
My simulations show that increasing D makes it progressively more difficult for the spins to rotate away from the x axis. At small D , the spins have more freedom to rotate continuously during magnetization reversal, while at larger D the two directions $+x$ and $-x$ become increasingly favored.



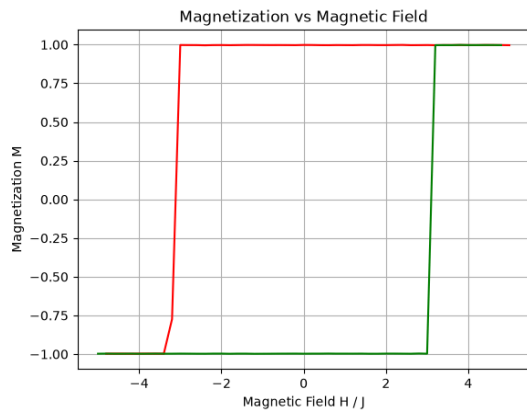
Anisotropy Term: 1



Anisotropy Term: 5



Anisotropy Term: 20



Anisotropy Term: 100

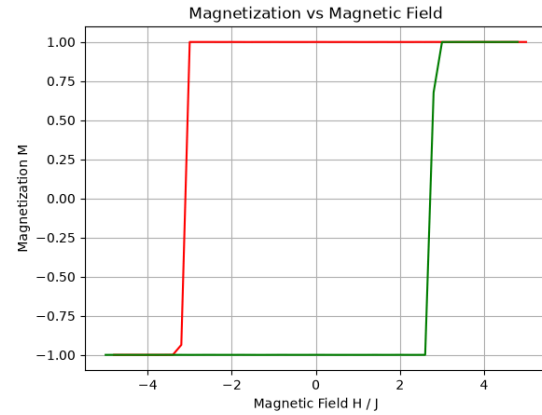


Figure3: XY-model hysteresis loops for several anisotropy strengths D using the same initial conditions and sweep conditions. D values $\rightarrow 0, 0.25, 1, 5, 20, 100$

As D is increased, the hysteresis loop initially becomes wider because the stronger anisotropy makes it more difficult for the spins to rotate away from the x -axis. However, I noticed that this increase does not continue indefinitely. At larger values of D , the hysteresis curves begin to change very little and approach a limiting shape. This occurs because once the anisotropy is sufficiently strong, the spins are already confined almost entirely to the $+x$ and $-x$ directions. Increasing D further therefore has little additional effect on the allowed spin orientations. In this regime, the XY model behaves increasingly like the Ising model, where the spins are restricted to only two directions. The coercive field therefore approaches a limiting value rather than continuing to increase without bound.

To show this more clearly, I extracted the coercive field for each value of D and plotted H_c as a function of anisotropy strength.

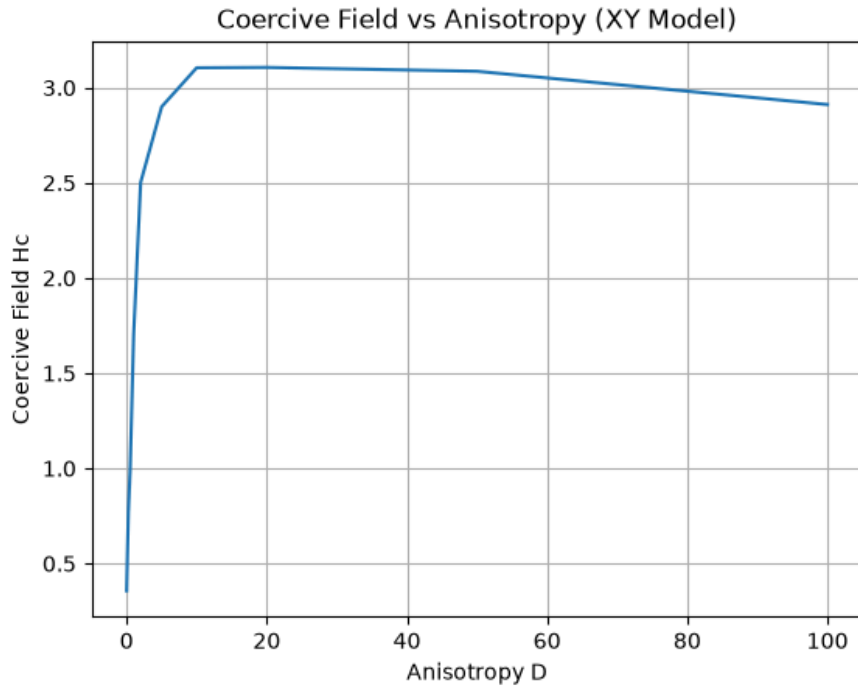


Figure4: XY model coercive field as a function of anisotropy strength D.

The coercive field increases rapidly at small D, reaching approximately $H_c \approx 3.1$, but then changes very little as D is increased further. At D = 100, the coercive field has decreased to approximately $H_c \approx 2.9$, which is very close to the value $H_c = 2.90$ obtained from my Ising simulation under the same conditions. This supports the expectation that the XY model approaches Ising-like behavior as the anisotropy becomes very large.

The XY coercive field slightly exceeds the Ising value at intermediate values of D. One possible reason is that the Monte Carlo dynamics of the two models are not identical. In the Ising model, a spin can only flip directly between +1 and -1, while in the XY model the spin changes through proposed angular rotations. Since hysteresis and the measured coercive field depend not only on the energy landscape but also on the update rule, number of Monte Carlo sweeps and field step size, the H_c value calculated will not exactly match that of the Ising model. At sufficiently large D, however, the spins become increasingly restricted to the $\pm x$ directions and the coercive field approaches the Ising result.

4. Heisenberg Model Extension

The Heisenberg model extends the XY model by allowing each spin to point in any direction in three dimensions. A unit spin can be represented using polar and azimuthal angles as:

$$S_i = (\sin \theta_i \cos \varphi_i, \sin \theta_i \sin \varphi_i, \cos \theta_i)$$

Using an easy axis and applied field along x, the Hamiltonian has the form:

$$E = -J \sum_{\langle i,j \rangle} S_i \cdot S_j - D \sum_i (S_i^x)^2 - H \sum_i S_i^x$$

The exchange term now uses the full three-dimensional dot product between neighboring spins. The anisotropy term penalizes orientations away from the x axis without distinguishing between +x and -x, while the external field selects which of those two directions is energetically favorable.

I have implemented the basic Heisenberg update and have been testing the spin proposal method to avoid introducing a preferred rotation direction numerically. Because a three-dimensional spin has more possible directions than an XY spin, extra care is needed to sample proposed orientations without bias. To verify the behavior of the Heisenberg implementation, I examined the individual spin components after running the system with Initial conditions: J=1, T=10, D=100, H=0.

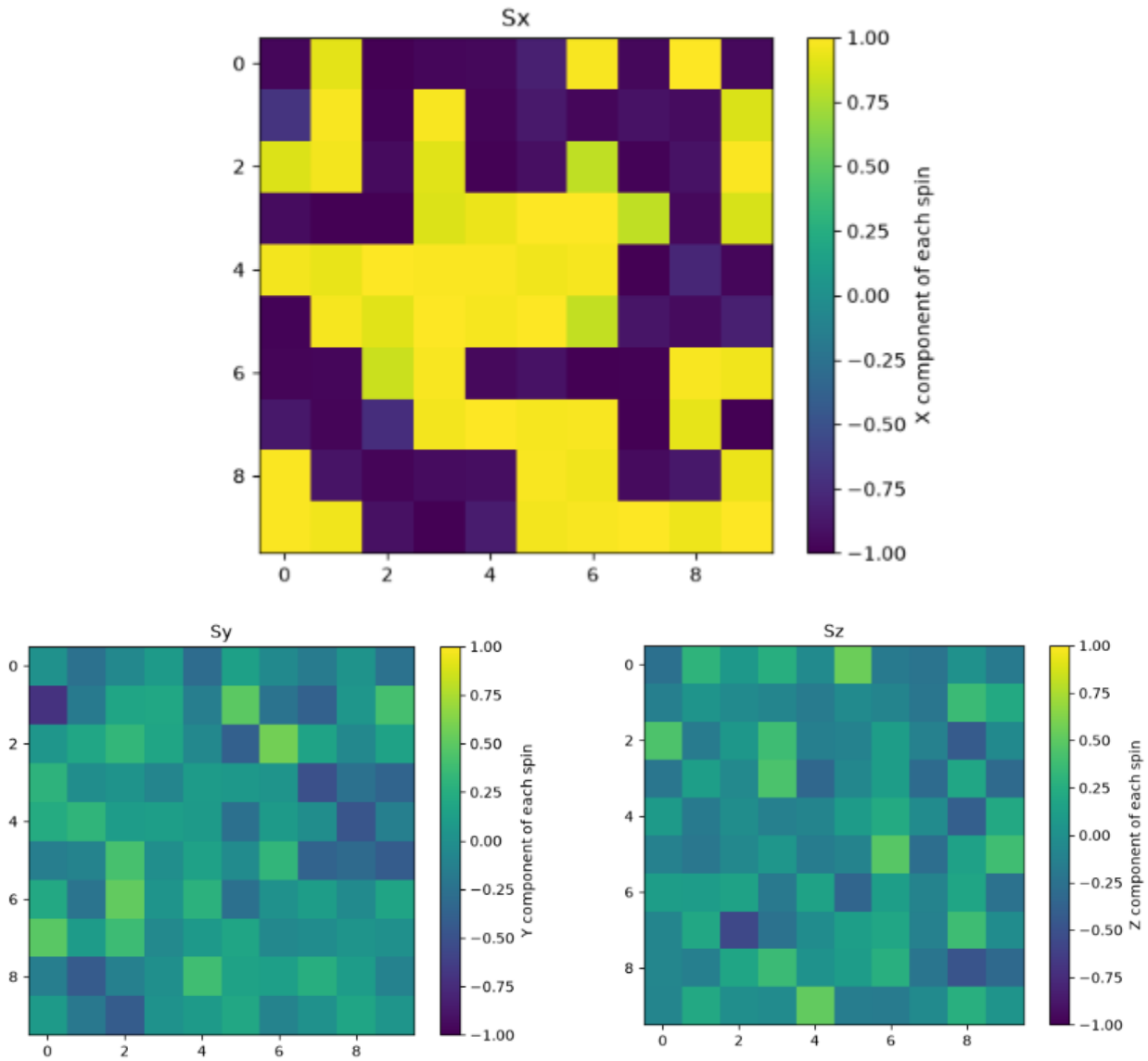


Figure5: color gradient for each component of spin in a 10x10 lattice in the Heisenberg Model after 1000 Monte Carlo sweeps using initial conditions: $J=1$, $T=10$, $D=100$, $H=0$.

With the large anisotropy $D = 100$, the x-components of the spins are concentrated close to either $+1$ or -1 , while the y and z-components remain close to zero. This shows that although the Heisenberg spins are free to point in any direction in three dimensions, strong anisotropy confines them close to the preferred x-axis. Since $H = 0$, the anisotropy does not distinguish between the $+x$ and $-x$ directions, so both orientations are present in the lattice. As D becomes sufficiently large, the Heisenberg model therefore begins to resemble the two-state behavior of the Ising model.

5. Heisenberg Model Hysteresis

When $D = 0$, the XY and Heisenberg spins can rotate freely within their allowed spaces but as D increases, deviations away from the easy axis occur less often as it costs more energy. In the large D limit, the low-energy orientations are concentrated near $+x$ and $-x$, so the models become increasingly similar to an Ising system with two allowed spin directions.

The simulations currently reproduce this qualitative trend: larger D gives stronger resistance to field-driven reversal and more Ising like hysteresis.

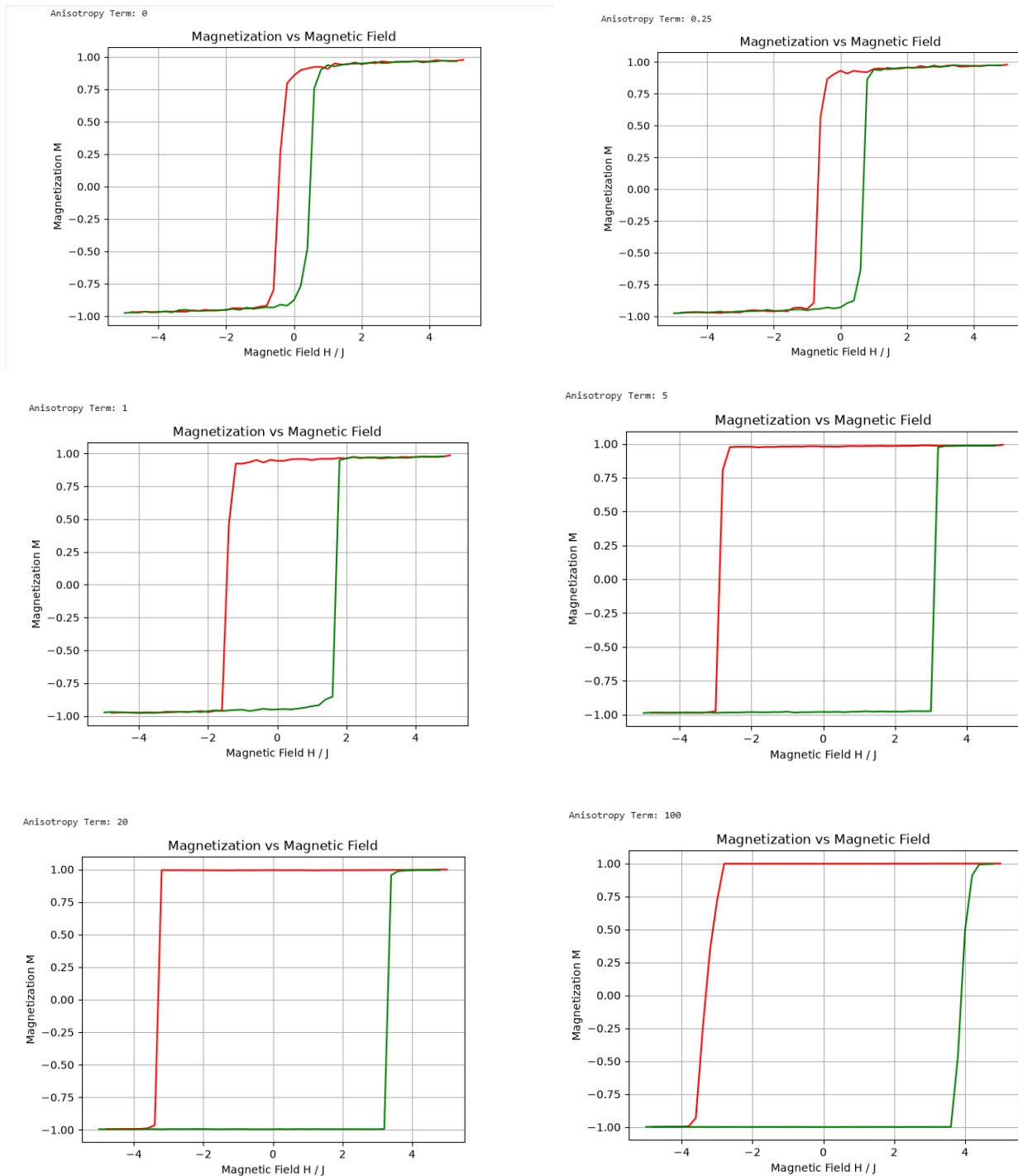


Figure6: Heisenberg model hysteresis loops for several anisotropy strengths D using the same initial conditions and sweep conditions. D values $\rightarrow 0, 0.25, 1, 5, 20, 100$

The Heisenberg model follows the same overall trend observed in the XY model. At small D , the spins have more freedom to rotate away from the easy axis, so the hysteresis loop is relatively narrow and the coercive field is small. As D is increased, the loops become wider and more square because the spins are increasingly confined toward the $\pm x$ directions. At sufficiently large D , further increases in anisotropy produce only small changes in the loop shape, indicating that the system is again approaching a strongly anisotropic limit.

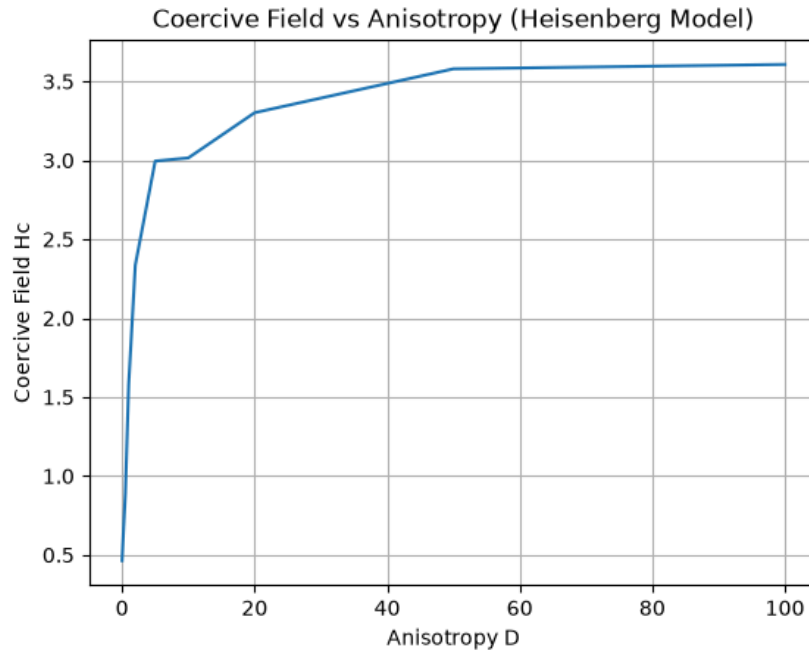


Figure7: Heisenberg model coercive field as a function of anisotropy strength D.

However, unlike the XY model, the coercive field in the Heisenberg simulation approaches a higher value. In my simulations, the large D Heisenberg coercive field approaches approximately $H_c \approx 3.6$, compared with approximately $H_c \approx 2.9$ for the XY model and the Ising simulation. Since both the XY and Heisenberg Hamiltonians should become increasingly Ising-like at very large D, this difference is likely related to the Monte Carlo dynamics and the way new spin orientations are proposed rather than representing a fundamentally different Ising limit. The Heisenberg model has an additional rotational degree of freedom, and its update procedure samples 3D orientations, so the rate at which the system escapes a metastable magnetized state can differ from the XY and discrete Ising models.

6. Limitations and Future Opportunities

One limitation of the current XY and Heisenberg simulations is the way new spin orientations are proposed during each Monte Carlo update. At the moment, a proposed update can rotate a spin into any orientation, potentially allowing it to move almost directly from one side of the easy axis to the other in a single step. Although the Metropolis acceptance rule still determines whether the move is energetically allowed, this update procedure may affect the simulated magnetization-reversal dynamics and therefore the measured coercive field.

A possible improvement would be to restrict each proposed update to a smaller angular change around the spin's current orientation. This would make the magnetization reverse through a sequence of smaller rotations rather than allowing very large changes in direction in a single Monte Carlo step. I could then compare several maximum proposal angles to determine how strongly the hysteresis curves and coercive fields depend on the Monte Carlo update size.

This may be particularly important when comparing the XY and Heisenberg models with the Ising model. In the Ising model, each spin has only two possible states and can only switch directly between them, whereas the continuous spin models have many possible intermediate orientations. The different update dynamics may therefore contribute to the differences I observed in the large D coercive fields, even though all three models are expected to become increasingly similar as the anisotropy becomes very strong.

Another future improvement would be to test whether the results converge with the number of Monte Carlo sweeps performed at each magnetic field value. Because the hysteresis simulation is not an equilibrium measurement, changing the number of sweeps effectively changes how much time the system has to escape

from a metastable state before the field is changed again. The field step size, lattice size, and proposed angular step can therefore all influence the measured coercive field.

For the Heisenberg model, the method used to generate new 3D spin orientations should also be thought up carefully. If completely new orientations are proposed, they should be sampled uniformly over the surface of a sphere rather than uniformly in angles, since uniform sampling of an angle does not correspond to uniform sampling of directions. Alternatively, small rotations around the current spin direction could be used instead.

These tests would help separate features caused by the Hamiltonian itself from features caused by the numerical Monte Carlo dynamics and determine whether the XY and Heisenberg models approach the Ising result more closely under comparable update conditions.

7. Machine Learning / AI Literature Survey

Machine learning (ML) and AI are increasingly being used to accelerate the search for new magnetic materials. Normally, what materials should be used for permanent magnets relies heavily on experiments and computational calculations such as density functional theory (DFT). These methods are still important because they provide reliable information about a material, but it can be computationally expensive or experimentally impractical to apply them to every possible composition and crystal structure. ML provides a way to screen much larger materials spaces and identify the candidates that are most worth investigating in more detail. ML has been applied to magnetic materials to predict properties such as saturation magnetization, Curie temperature, magnetic anisotropy, and thermodynamic stability.

Rather than performing expensive calculations on every hypothetical material, an ML model can first use information from previously calculated or experimentally measured materials to estimate which candidates are likely to satisfy these requirements. The promising candidates can then be passed to more accurate DFT calculations for validation. In this way, ML acts as a sort of fast screening tool for the various possible candidates.

Mal and Sen demonstrated this approach using existing rare-earth free magnet databases where they trained neural network models to predict quantities such as heat of formation and saturation magnetization, while also classifying materials according to stability and magnetic anisotropy [3]. The trained models were then used to screen 686 rare-earth free materials. After the ML screening, calculations were performed on the promising candidates, resulting in 21 proposed rare-earth free permanent-magnet materials. This shows how ML can reduce a large search space before more computationally expensive calculations can be performed later.

A more recent approach combines several computational methods into a larger AI assisted workflow. Chelikowsky et al. describe a framework combining ML, adaptive genetic algorithms, and DFT to search for rare-earth free Fe-Co-X compounds, where X includes B, C, P, Si, and S [2]. Fe-Co based materials are attractive to researchers because Fe and Co can provide high magnetization, but their normally high symmetry structures often have weak magnetic anisotropy. Adding other elements and searching for lower symmetry crystal structures provides a possible way of increasing the anisotropy while retaining high magnetization.

In this approach, ML is used to rapidly screen very large numbers of hypothetical structures, while adaptive genetic algorithms search for favorable crystal structures and DFT is then used to calculate various properties. One example described in the paper involved screening more than 350,000 hypothetical Fe-Co-B structures using a trained ML model before reducing the search to a much smaller number of candidates for DFT calculations. As an example of this model working, this process identified Fe₃CoB₂ as a promising rare-earth free magnet candidate with both high magnetization and strong uniaxial anisotropy.

Overall, these studies show that one of the main changes introduced by AI and ML is a shift from testing relatively small numbers of materials individually toward computationally screening very large possible compositions and structures first. The general workflow with the help of AI can be summarized as:

Existing materials data → ML model → large scale candidate screening → promising candidates → DFT / more detailed calculations → experimental candidates

This allows computational effort to be concentrated on materials that already have a higher predicted chance of containing the properties needed for a permanent magnet.

8. Limitations and Future Use of ML

Although ML can greatly accelerate materials screening, there are still limitations. One of the largest problems in magnetic material research is the amount and quality of available data. Compared with many other applications of ML, datasets containing reliable magnetic properties can be relatively small, and some important quantities such as magnetocrystalline anisotropy may only be available for a fraction of known materials. This makes it more difficult to train models that remain accurate when predicting materials very different from those contained in the training data. The broader review by Jiang et al. identifies data scarcity and physical interpretability as major challenges and discusses physics informed machine learning as one possible way of improving predictions when data are limited [1].

Because of this, the most useful role of ML currently uses it in an existing workflow rather than as a replacement for DFT or experiments. ML can search a much larger materials space quickly, while calculations can verify whether the predicted structures are stable and whether their magnetic properties are physically reasonable. Future approaches may increasingly use active learning methods, where the results of new calculations or experiments are added back into the ML model so that it can improve its predictions and decide which materials should be investigated next.

References

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